PRODUCT, WORK AND EVIDENCE

GEO CRM

An engineering case record

Sales, stock, expense requests and 13-week liquidity

One evolving local application, continued across agent environments.
A current English demonstrator, reconstructed history and fresh checks.

01 **Sales and warehouse**
Original Codex implementation

02 **Expense requests**
DeepSeek Harness continuation

03 **LIQ-13 and current presentation**
Return to Codex; English / Graphite

APPLICATION	1.3.1-demo / source identified by SHA-256
VERIFICATION	36 tests; 182 daily rows; 26 weekly aggregates
PREPARED	5 November 2026 / International English

ENGINEERING RECORD

What this report establishes

GEO CRM is a working local demonstrator for a fictional surveying-equipment supplier. It combines sales, stock handling, quotations, expense requests and a 13-week cash-flow model. This report describes the current English application, then connects its behaviour to fresh checks and recoverable historical evidence.

Distinction	Meaning in this report
Implemented	Present in the identified source and data model. Presence alone is not a successful execution.
Verified now	Executed for this report against an exact source copy, real PostgreSQL and synthetic data.
Retained evidence	A preserved record of an earlier execution. It is attributed to its own date and version.
Human judgement	A recorded user decision or observation. It is not inferred from an agent test result.

The current result

The identified application is 1.3.1-demo: English-only, GBP, fictional UK companies and a shared Graphite theme. The existing acceptance suite passed 36 tests. A separate case exercised the customer-to-shipment path and the expense-to-forecast connection. An independent calculation reproduced 182 daily rows and 26 weekly aggregates.

The report also separates a positive weekly closing balance from a negative balance earlier in that week. A 30-day receipt delay produces a minimum of minus GBP 10,100 in the illustrated case. This is a conditional model result, not a probability estimate or a prediction of actual insolvency.

Reading route. Sections 1-3 explain the product and work. Figures 1-12 show the case. The evidence matrix and appendices support reuse by a site editor without reconstructing the chat history.

ENGINEERING RECORD

Contents

Text pages use A4. Complete interface plates use A3 landscape so dense working screens remain readable; evidence tables use A4 landscape. Original JPEGs are included alongside the PDF.

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ENGINEERING RECORD

1. The assignment and the resulting product

The initial request was, in practical terms: build a compact CRM demo for a company selling surveying equipment; learn from Bitrix24, choose the implementation, make the workflows real and verify them locally. The human delegated engineering execution in S4: independent work within agreed limits.

The preserved brief was more specific than that short summary. It required Python and PostgreSQL, one sales pipeline, one warehouse, synthetic data, roles, stock reservations, saved quotation versions and a verification plan before the main implementation. No implementation design or ready set of UI screenshots was supplied. The agent had to research the interface and turn the text into a coherent system.

Stage	Delivered capability	Evidence status
31 Oct 2026 / initial Codex work	Sales, customers, activities, stock, quotations, history and management overview.	Retained original brief, architecture, V&V and chronology.
2 Nov / DeepSeek Harness	Expense request entity, roles, approval threshold and payment calendar.	Retained implementation trace and answers; current code and new execution.
2 Nov / return to Codex	LIQ-13 method and application module, saved forecasts and later UI work.	Retained method activation, model and verification material.
5 Nov / localisation and theme	English synthetic data, GBP, English documents and Graphite across the application.	Current 1.3.1 source identity and new report checks.

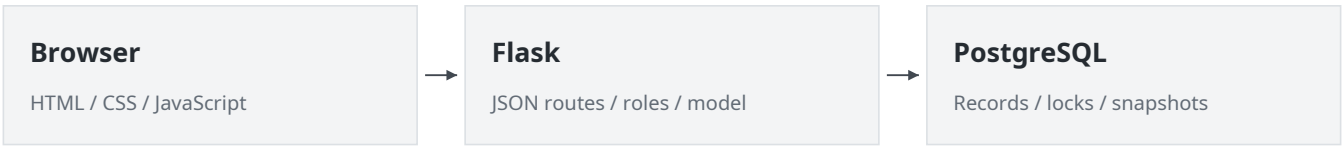
The current product is therefore larger than the original 36-minute build. The finance module, liquidity model, documentation, English conversion and later visual work must not be attributed to that initial interval.

Project boundary. This is the GEO CRM project in the present iEWR instance. Its LIQ-13 module is not the separate FIN-LIQ-13 experiment in an iEWR-7.0.1-DMG-TEST project. No material from that separate project was used. The retained Bitrix24 research and reference-image hashes are indexed in evidence/REFERENCE_REGISTER_EN.md.

ENGINEERING RECORD

2. How the system fits together

The implementation is a small Flask monolith. A browser loads HTML, CSS and plain JavaScript; JSON requests reach Python routes; PostgreSQL stores the records and enforces key constraints. There is no frontend build pipeline, microservice topology or external service required for the local demo.



Sales + stock + expenses feed the local workflow; dated cash inputs feed LIQ-13.

Data group	Tables / responsibility
Identity and installation (2)	users; schema_migrations. Six roles, seven demo accounts.
Sales and customer work (7)	companies, contacts, deals, deal_items, activities, comments, quotes.
Stock (4)	products, reservations, stock_movements, stock_operations.
Audit (1)	history, with actor, time, entity and before/after data.
Expense requests (2)	expense_articles; payment_requests.
Liquidity (3)	liquidity_settings; liquidity_items; liquidity_runs.

The model contains 19 tables: the original 14 plus two expense tables and three liquidity tables. Before this report, the working demo contained 23 companies, 35 contacts, 28 products, 64 deals, 86 activities, 13 expense requests and two saved forecasts. These are synthetic records, not evidence of commercial use.

Money uses PostgreSQL numeric columns and Python Decimal arithmetic. Reservations and shipment use transactions and row locks. Quotation versions and saved forecasts carry independent snapshots; immutable-record triggers protect those stored versions and the audit history. Protection does not make the underlying business assumptions correct.

ENGINEERING RECORD

3. What the human and the agents contributed

Decision / work	Responsibility and recoverable basis
Purpose, scope and authority	The human requested the local demo, supplied the business context and explicitly delegated S4 execution.
Research and implementation design	The initial agent selected official Bitrix24 references, Flask/plain JavaScript, database entities, transactional stock handling and snapshot-based documents.
Verification organisation	The agent wrote the V&V plan before the main implementation, then combined real HTTP/PostgreSQL tests with browser scenarios and a restart check.
Finance policy	The human chose a separate expense entity, core scope, the original threshold and 10:00 payment-day cut-off, links instead of uploads, and manual handling without accounting/bank integration.
Liquidity formation	The human asked for a method before implementation. The agent constructed LIQ-13 from adequate general and professional foundations, added the local method and implemented it.
Presentation and later direction	The human selected the English-only direction and Graphite palette. The agent carried these through the existing system and verified the changes.

Work formed around results

The original execution had an architecture and a V&V plan, but was not a fully enumerated script. Preserved progress records show feedback changing the work: an empty owner filter prevented the first kanban load; browser execution exposed it and led to a correction. A rendered PDF revealed a missing currency glyph; the document treatment was adjusted and checked again. A form later demanded a contact although the server allowed a company-only deal; the mismatch was corrected.

These are recorded examples of organising the next necessary work after obtaining new information. They are not a measurement of how another model would have behaved. The present report did not replay those original defects.

Human assessment. The user reported that the project started naturally after opening it in DeepSeek Harness. The user also chose policy and visual directions. No retained formal user acceptance establishes all business properties or production readiness.

ENGINEERING RECORD

Methods, continuity and the measured intervals

iEWR supplied entry instructions, source priority, a method repertoire and continuation material. FPF served as the pinned general methodological source; MethodPack held local reusable methods. DMG denotes finding, reusing, adapting or constructing an adequate method. The historical DeepSeek record shows recovery from CURRENT, handoff, the original brief, code and tests before the first implementation edit. The extension reused the same application and database rather than creating an independent product.

The LIQ-13 basis records use of FPF B.5.FM and C.29 for framing the model and balancing flows, with C.39/C.41 reused as adequate foundations. The local addition specifies concrete cash dates, eligible request states, original-date delay rules, daily calculation, weekly views, duplicate-source detection and reproducible scenarios. It remains a candidate method with bounded verification, not a validated universal financial standard.

For this report, the adequate C.37/C.41 basis was reused for representations and evidence scope, together with local LIQ-13 and established software verification practice. No new method or MethodPack activation was necessary. The current MethodPack consistency check passed; it checks file/description consistency, not the effectiveness of a method.

Observed interval	What it measures	What it does not measure
36 min 11 s	One original Codex implementation turn: 2,171,030 ms in the retained chronology.	The cost or total development time of the current multi-module product.
11 min 51 s	DeepSeek implementation turn 5: 711.026 s.	The preceding discussion, the whole transfer experiment or this later verification.
Not established	Token consumption, monetary cost and a controlled run without iEWR.	No speed-up factor or causal advantage can be calculated from these records.

The DeepSeek alias recorded in the session header is deepseek-flash, reasoning high; Harness Desktop is 0.2.1-rc.5. The retained project evidence does not establish the exact Codex model IDs. The current interface change cannot retrospectively identify an earlier model.

Reuse without ritual. No additional FPF/MethodPack reads were found in the DeepSeek export. That alone does not demonstrate failure of iEWR: sufficient project practice and professional competence can be reused. The trace neither establishes a new source-selection step nor warrants inventing one.

ENGINEERING RECORD

4. A customer-to-shipment case

The new case was executed on a restored copy of the English demo. It created Meridian Field Services Ltd, contact Jordan Ellis and deal 65, Survey kit for the Northbank site. The manager, warehouse, managing director, finance director and accountant used separate demo identities for their respective operations.

Equipment	Quantity	Unit price	Line amount
Terrain M4 GNSS receiver - case unit	1	GBP 2,650.00	GBP 2,650.00
Fieldline HD survey tripod - case unit	1	GBP 189.00	GBP 189.00
Total			GBP 2,839.00

Warehouse staff created the two case catalogue records and received two units of each with explicit operation keys. The manager created the customer, contact, deal, line items, a call activity and quotation 65-1. The quotation remained unchanged during a temporary deal-price edit; the original deal prices were then restored before fulfilment.

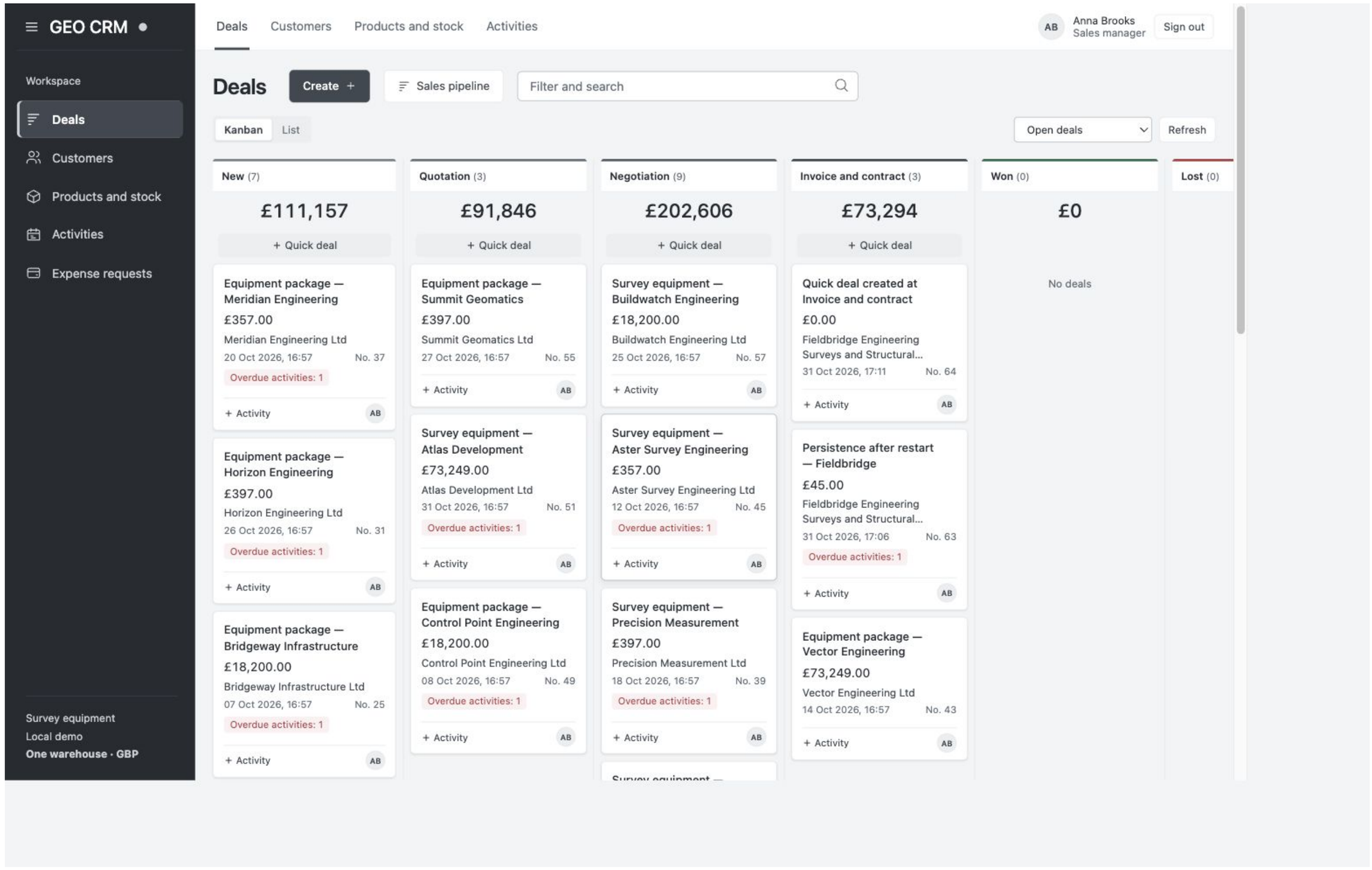
State of each case product	On hand	Reserved	Available
After receipt	2	0	2
After reservation	2	1	1
After shipment	1	0	1
After replaying shipment	1	0	1

A manager shipment attempt was denied. Warehouse shipment completed the deal as won, removed its reservation and changed stock consistently. A repeated reserve did not duplicate the reservation; a repeated shipment did not deduct stock again. A different manager was denied direct access to the deal. These observations complement the existing concurrency and atomicity tests.

Evidence route. C02-C05 in the claims matrix; scenario.json, scenario-actions.json and case-quotation.pdf. The figures show the completed records. Creation and stock transitions were executed by HTTP; the browser observations are identified separately.

PRODUCT IN USE

Figure 1. The sales workspace

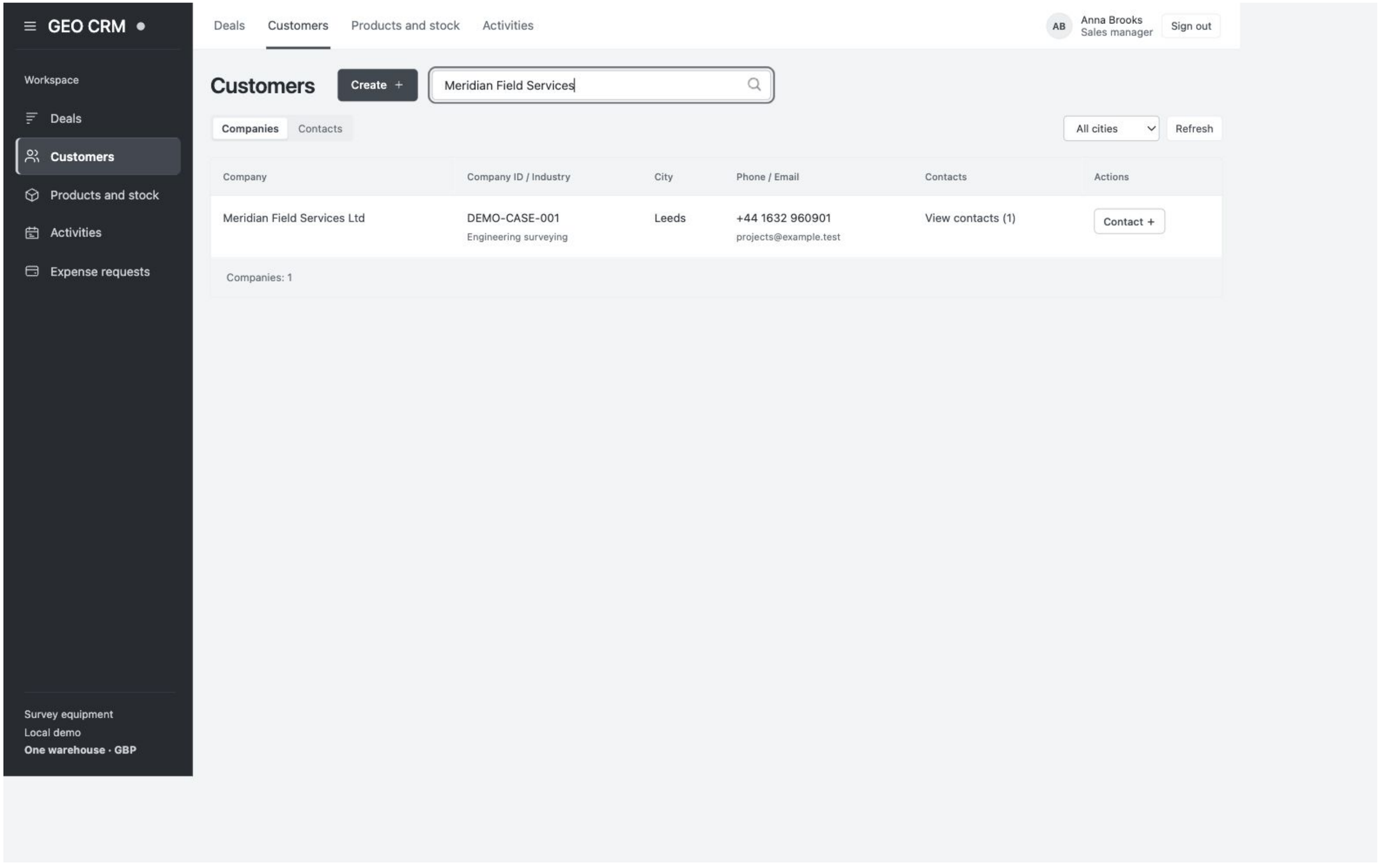


Role: manager / C01

Sales manager: open deals in the Graphite kanban; existing seeded records provide context. The completed Northbank case is shown separately. A real manager view of the seeded open pipeline. Kanban and list are two presentations of the same deal records. This is the current Graphite implementation, not an image of Bitrix24 or a UI mockup.

PRODUCT IN USE

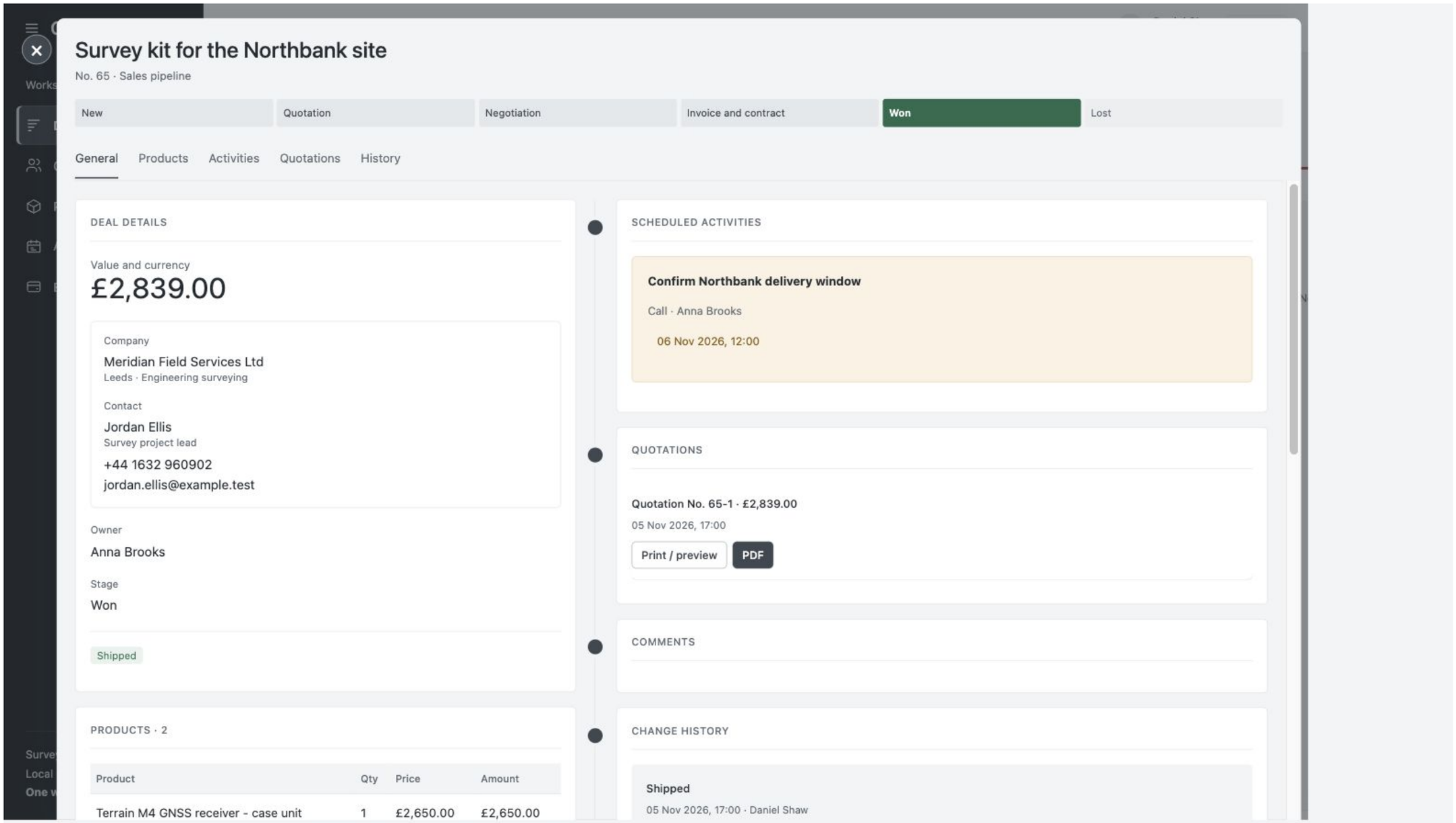
Figure 2. The case customer



Role: manager / C02
Sales manager: the new synthetic customer Meridian Field Services Ltd is found by search, with a linked contact and synthetic contact details. The search result identifies the new fictional customer. Its contact was created with the company relationship, and both are used by deal 65. Synthetic example.test addresses and demonstration phone numbers avoid real contact data.

PRODUCT IN USE

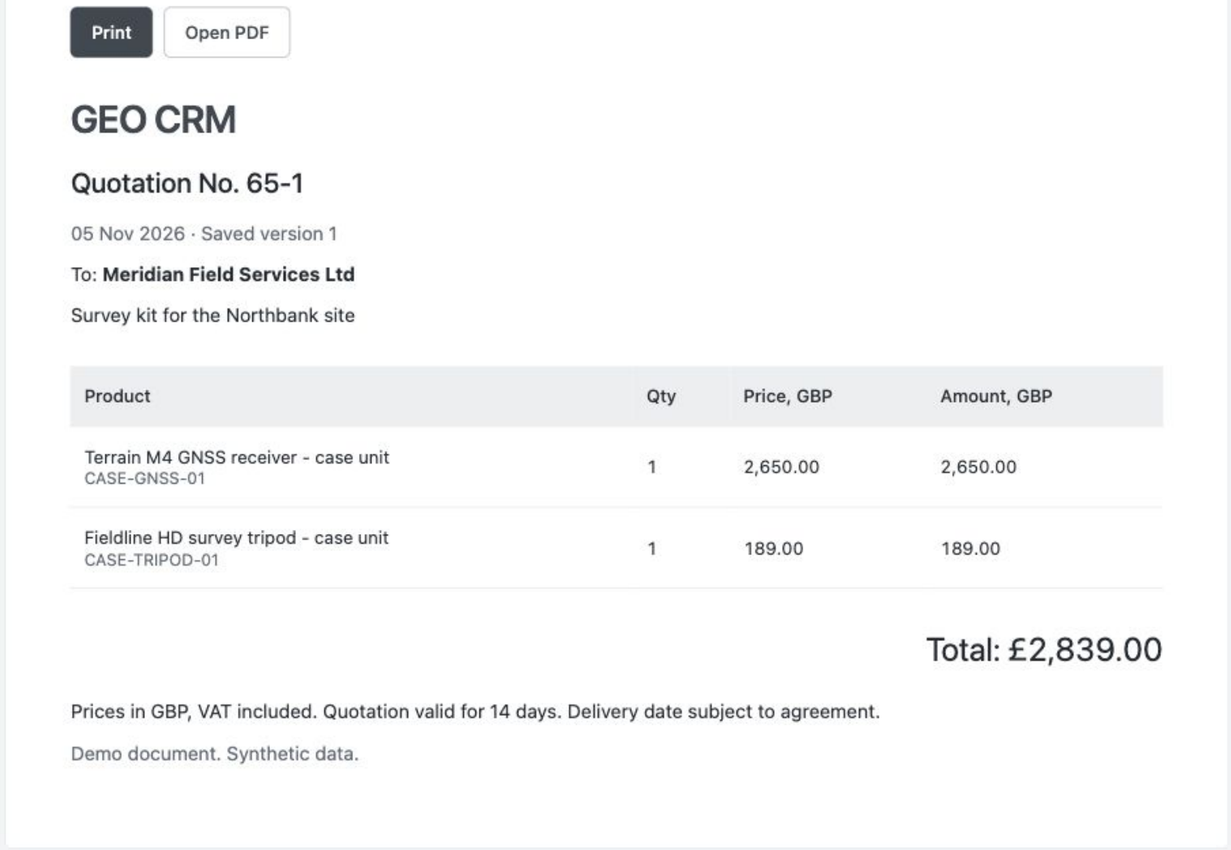
Figure 3. The completed deal



Role: warehouse / C03
Warehouse: deal 65 is Won and Shipped. The customer, contact, planned activity, quotation and shipment history identify the same case; the product panel reports two lines. Reservation release is verified by C04. The warehouse role can inspect the sales context needed for fulfilment. The shipped state, equipment lines and history belong to the same case; the role is not given a manager editing workflow here.

PRODUCT IN USE

Figure 4. A saved commercial document



Print Open PDF

GEO CRM

Quotation No. 65-1

05 Nov 2026 · Saved version 1

To: **Meridian Field Services Ltd**

Survey kit for the Northbank site

Product	Qty	Price, GBP	Amount, GBP
Terrain M4 GNSS receiver - case unit CASE-GNSS-01	1	2,650.00	2,650.00
Fieldline HD survey tripod - case unit CASE-TRIPOD-01	1	189.00	189.00

Total: £2,839.00

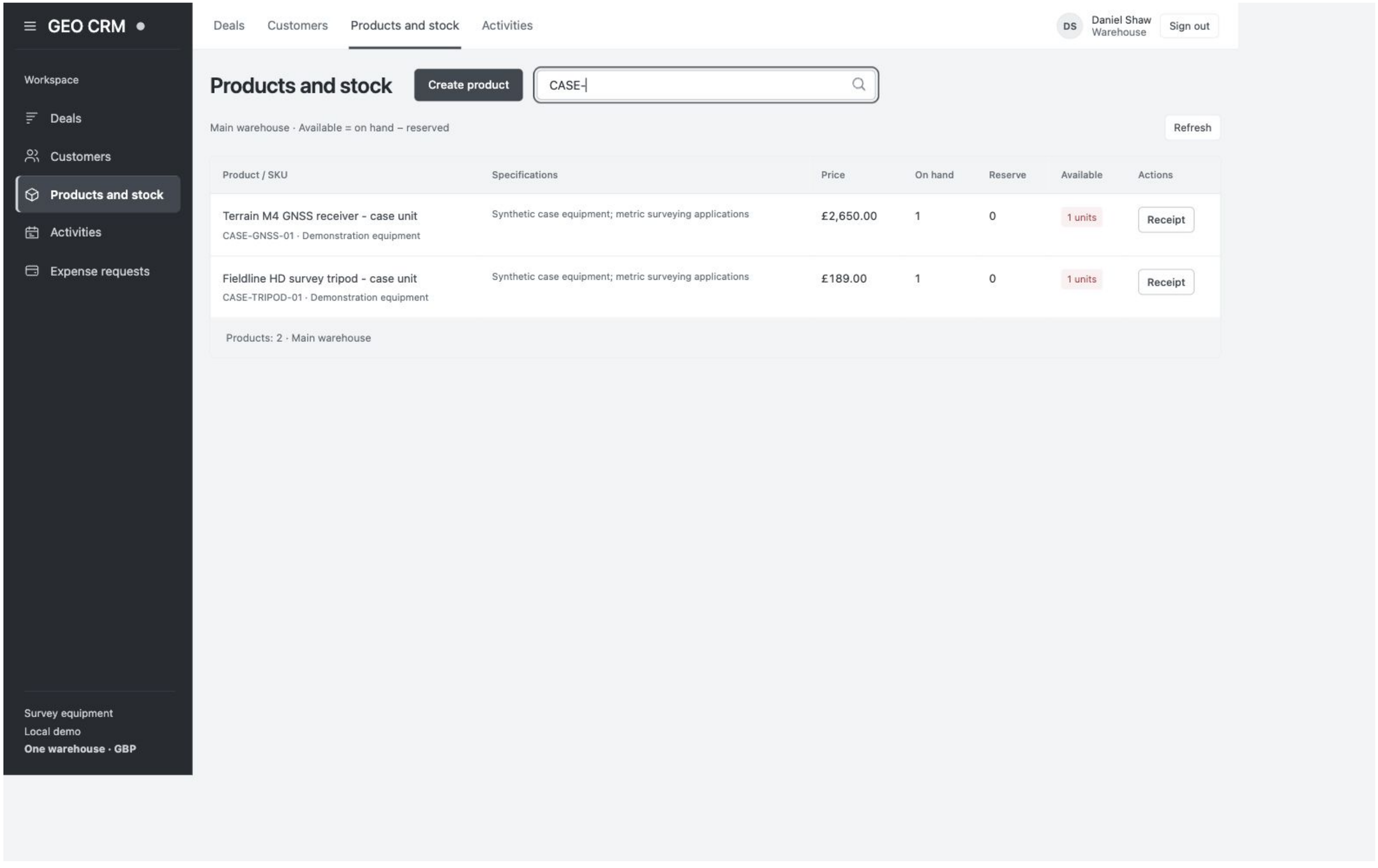
Prices in GBP, VAT included. Quotation valid for 14 days. Delivery date subject to agreement.

Demo document. Synthetic data.

Role: warehouse / C05
Saved quotation 65-1 totals GBP 2,839.00. A temporary change to deal prices and subsequent shipment left this stored document unchanged. Quotation 65-1 preserves its own line descriptions and prices. The HTTP check changed the deal price temporarily and compared the stored quotation before and after; a screenshot alone would not establish that property.

PRODUCT IN USE

Figure 5. The inventory consequence



Role: warehouse / C04
Warehouse: each case product now has one unit on hand, no reservation and one available unit, after receiving two and shipping one. Replaying shipment did not reduce stock again. The two case SKUs make the stock effect easy to trace. Two units were received and one shipped for each SKU. The quantities shown here were also checked after repeating the shipment request.

ENGINEERING RECORD

5. Expense requests: a separate workflow

Expenses are not represented as sales deals. A request has an amount, category, payee, requester, due date, optional deal reference, document link, approval markers and explicit status. Four category groups contain 14 seeded expense categories. The application does not upload the invoice or send a bank instruction.

Stage	Who acted in the case	Observed result
Draft	Sales manager Anna Brooks	Created ER-0014 for GBP 1,250, linked to deal 65.
Above-limit director decision	Managing director Emma Hart	Recorded the required director approval.
Approved	Finance director Irene Bell	Approval succeeded after the director decision; before it, HTTP 409.
Scheduled	Finance director Irene Bell	Payment date saved as Tuesday, 10 November 2026.
Paid	Accountant Olivia Turner	Recorded through the real UI after the forecast captures.

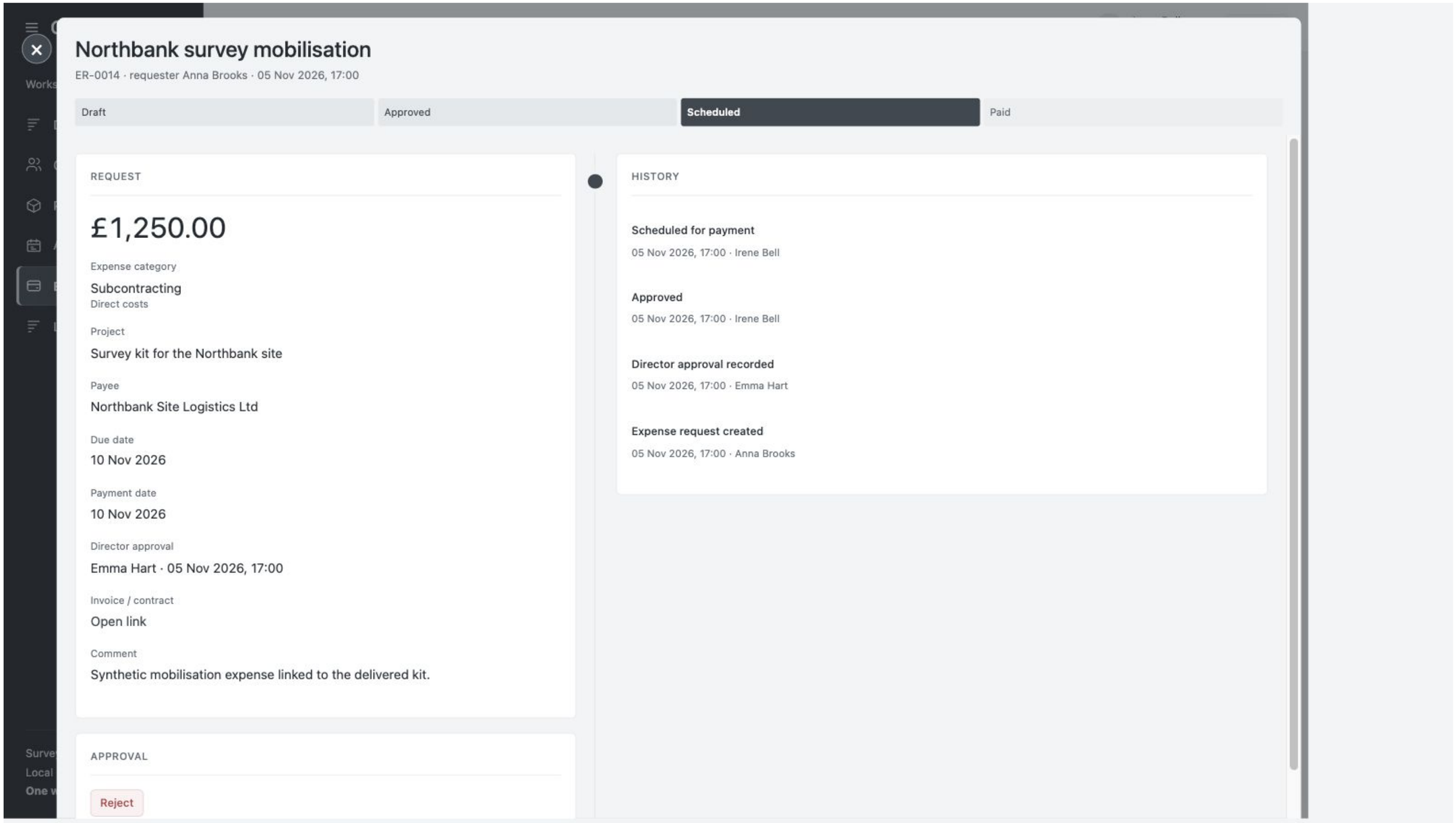
The current synthetic threshold is GBP 1,000. The earlier demo used RUB 100,000; the English dataset was rescaled 1:100, including this boundary. That was a synthetic redenomination, not an exchange-rate conversion. A manager attempt to approve this case request was denied by the server.

Ordinary payment days are Tuesdays; urgent requests can use Tuesday or Thursday. The selected cut-off is 10:00 on the payment day, using the local server clock. The saved payment date drives the scheduled commitment. The calendar schedules a date; it is not a bank-execution gate. The case was scheduled on 5 November for 10 November, then marked Paid on 5 November solely to exercise the synthetic recording and exclusion path.

What the workflow confirms. The illustrated path confirms the above-limit approval sequence, scheduling and manual payment recording in the stated roles. The separate limitations appendix records two remaining finance issues and an unresolved self-approval policy; this case does not erase them.

PRODUCT IN USE

Figure 6. The scheduled request



Role: finance director / C06
Finance director: request ER-0014 for GBP 1,250 is scheduled for 10 November, after managing-director approval. History identifies the actors and transitions. The finance director sees the GBP 1,250 request, its linked deal, payment date and the managing director approval. The history records four steps under three separate identities. This image was captured before payment was recorded.

ENGINEERING RECORD

6. What LIQ-13 models

LIQ-13 answers a concrete conditional question: given the opening cash, dated expected receipts and payments, what happens if customer receipts arrive later? The horizon is 91 days, presented as 13 seven-day weeks. It calculates daily closing balances and then derives weekly summaries.

Cash identity. Closing cash = opening cash + receipts due that day - payments due that day. Each day becomes the next day's opening cash. The minimum includes the initial opening balance and all daily closing balances.

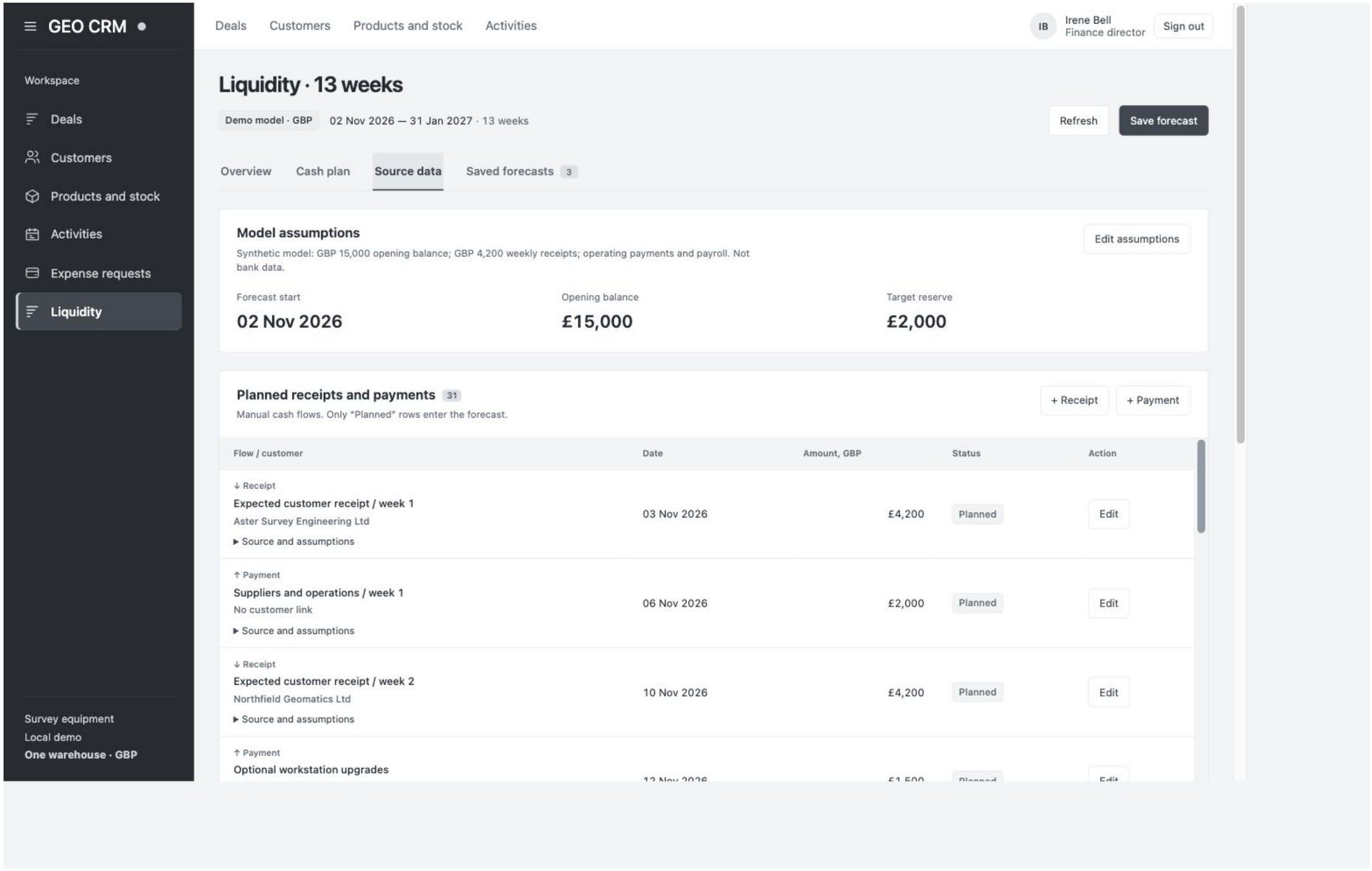
Input / rule	The illustrated case
Horizon	2 November 2026 to 31 January 2027; 91 days.
Opening cash / reserve	GBP 15,000 / GBP 2,000. These are model assumptions.
Manual plan	31 planned rows: the existing 30 plus a GBP 2,839 receipt for the delivered case kit on 12 November.
Open expense commitments	Approved/scheduled requests enter by the adapter rule; drafts are optional. The case request occurs exactly once among 37 total input sources.
Receipt stress	All customer receipts shift 30 calendar days from their original expected dates; payments remain on their dates.
Outside the horizon	Shifted receipts after day 91 are reported separately, not treated as lost receipts or silently dropped.

A deal total does not automatically become a cash receipt. The case explicitly adds a receipt linked to the customer and deal. This prevents a sales value from masquerading as a cash schedule. Manual payments can still duplicate expense commitments; the owner of the model must reconcile those inputs.

Additional implemented controls allow a selected-customer delay, receipt haircut, deferral of marked payments, optional inclusion of drafts and opening funding. The illustrated contrast varies only receipt timing. Open overdue flows are assumed on day one, after any scenario shift from their original dates. Paid and rejected requests are excluded. The model does not estimate probabilities, interest, foreign exchange or the order of transactions within a day.

PRODUCT IN USE

Figure 7. Assumptions and source rows



Role: finance director / C07

Finance director: opening cash GBP 15,000 and reserve GBP 2,000; manual receipts and expenses are explicit dated inputs. The case adds a linked GBP 2,839 expected receipt and the scheduled request. Opening cash, reserve and planned rows are visible and editable to the finance role. The captured scenario.json contains the exact input set used for the independent arithmetic check; these are synthetic assumptions, not a bank balance.

ENGINEERING RECORD

7. The cash effect and why daily detail matters

Same input set	Minimum	First cash gap	Funding to zero
Baseline / no receipt delay	GBP 8,850	None	GBP 0
All receipts delayed 30 days	-GBP 10,100	13 Nov 2026	GBP 10,100
Same delay + opening funding	GBP 0	None	GBP 10,100 injected

The delayed scenario needs GBP 12,100 to preserve the selected GBP 2,000 reserve, under the same linear assumptions. GBP 16,800 of expected receipts lie beyond the 91-day horizon. These are modelled requirements for this input set, not financing recommendations or authorised funding actions.

A weekly close can conceal a daily shortfall

Week 11: 11-17 January 2027	Daily closing cash
11-13 January	-GBP 61 on each day
14 January: receipt GBP 4,200	GBP 4,139
15 January: payment GBP 2,000	GBP 2,139
17 January: weekly closing balance	GBP 2,139

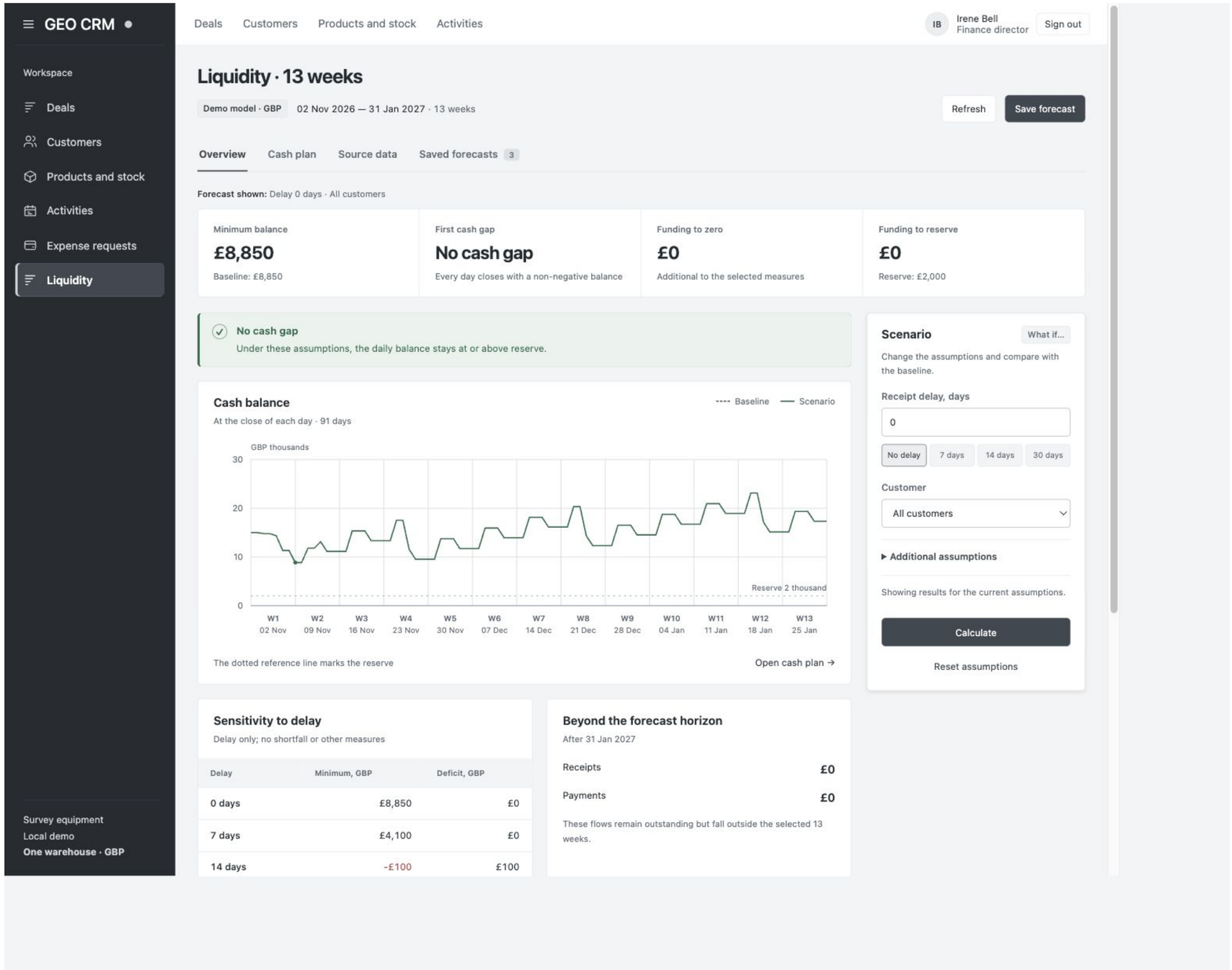
Week 11 closes positively, but the first three days close below zero. Week 13 has the same qualitative problem: a minimum of minus GBP 1,661 and a closing balance of GBP 539. Reading only the weekly closing row would conceal those modelled cash gaps.

The independent checker did not import the application engine. It reconstructed dated flows in integer pence, compared all 182 daily rows and 26 weekly aggregates, checked the minimum, first gap, funding and beyond-horizon totals, and verified the constant shift from opening funding across another 91 daily rows.

Different from the untouched demo. The working demo without the new case has baseline GBP 7,950 and 30-day minimum -GBP 8,850. The figures in this report use the new case inputs, so their GBP 8,850 / -GBP 10,100 results are intentionally different.

PRODUCT IN USE

Figure 8. The baseline

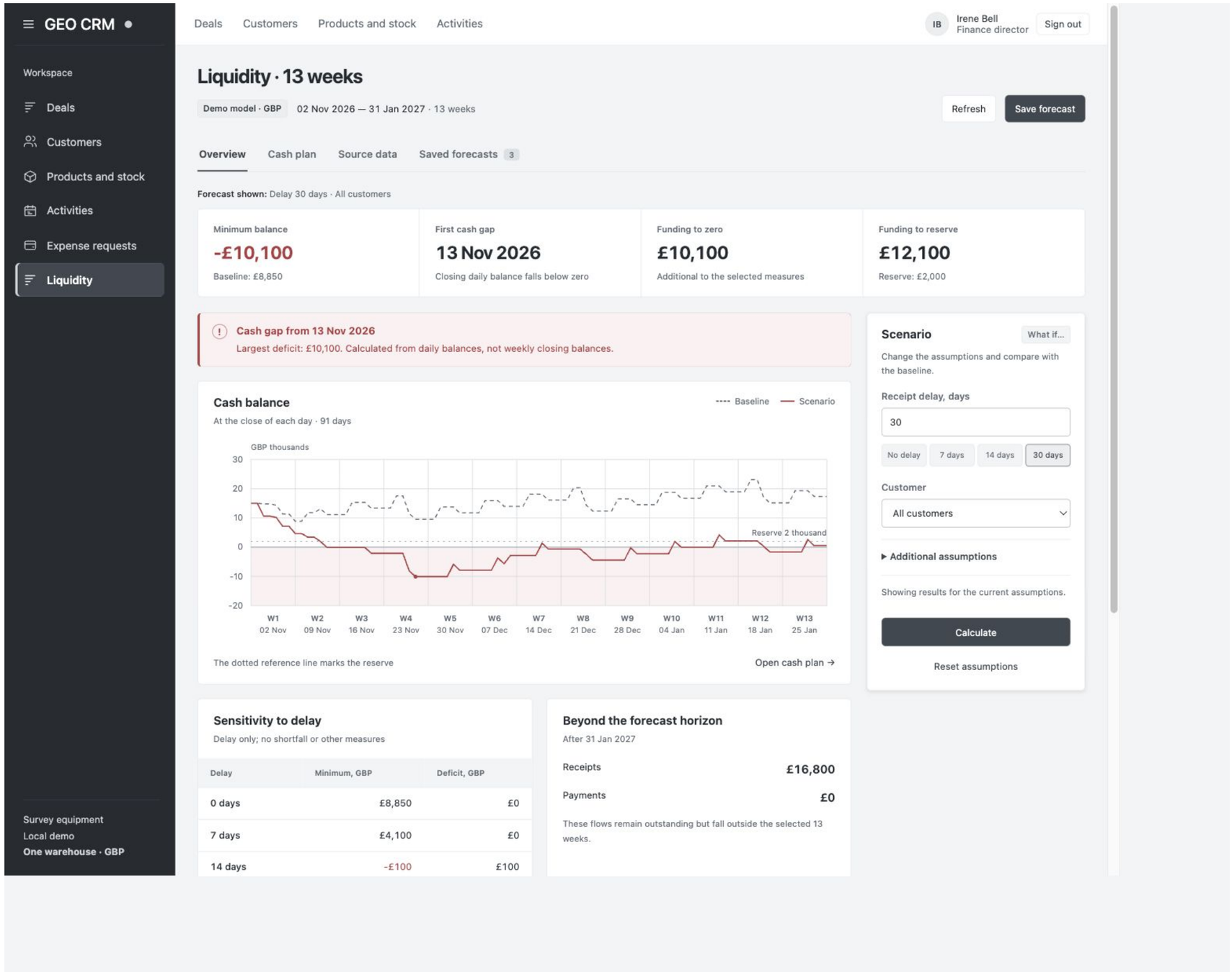


Role: finance director / C08

Finance director: the case baseline has no cash gap and a minimum daily closing balance of GBP 8,850. Zero delay matches the baseline exactly. Zero delay gives the same daily trajectory as the baseline and no cash gap. The displayed GBP 8,850 minimum is from the disposable case with its additional planned receipt and expense, not from the untouched main demo.

PRODUCT IN USE

Figure 9. A controlled receipt delay



Role: finance director / C09

Finance director: delaying all customer receipts by 30 days produces a GBP 10,100 deficit, first seen on 13 November. GBP 16,800 of receipts move beyond the horizon. Only the receipt timing assumption changes. The minimum falls to minus GBP 10,100, first negative on 13 November. The numerical check uses the full input and daily arrays, not the appearance of the plotted line.

PRODUCT IN USE

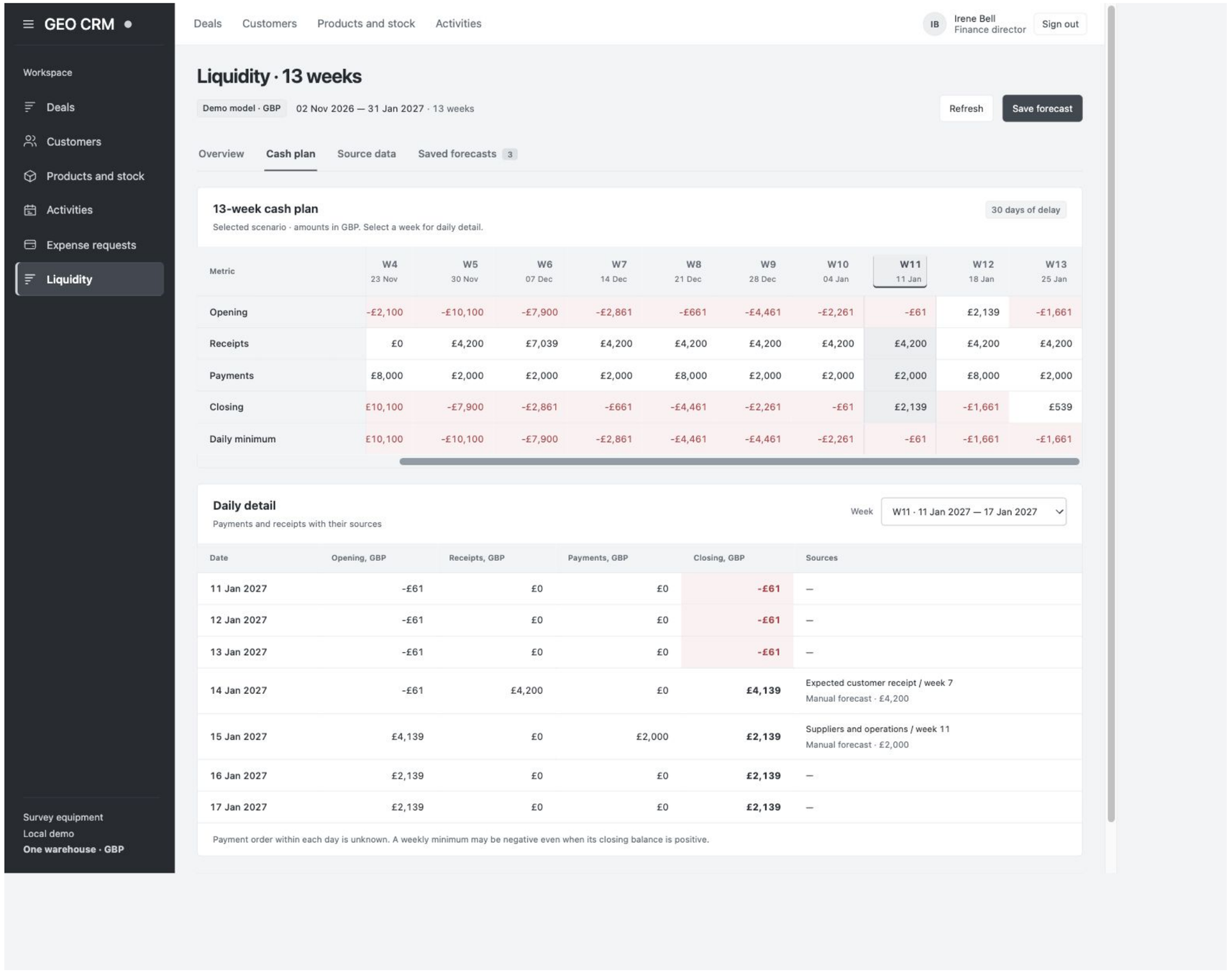
Chart detail. The saved stress scenario



Role: accountant / C09
Accountant: native chart close-up of saved forecast 3, preserving the same input set and 30-day result shown in Figure 9. Axes are in GBP thousands; the baseline is dashed and the scenario solid. This native close-up preserves the axes and all 13 week labels at a readable scale. It was captured directly from saved forecast 3 under the accountant role. Its inputs and result are identical to Figure 9; the data were not replaced or redrawn. The baseline is dashed and the scenario is solid; the shaded area marks negative cash.

PRODUCT IN USE

Figure 10. A positive week with negative days

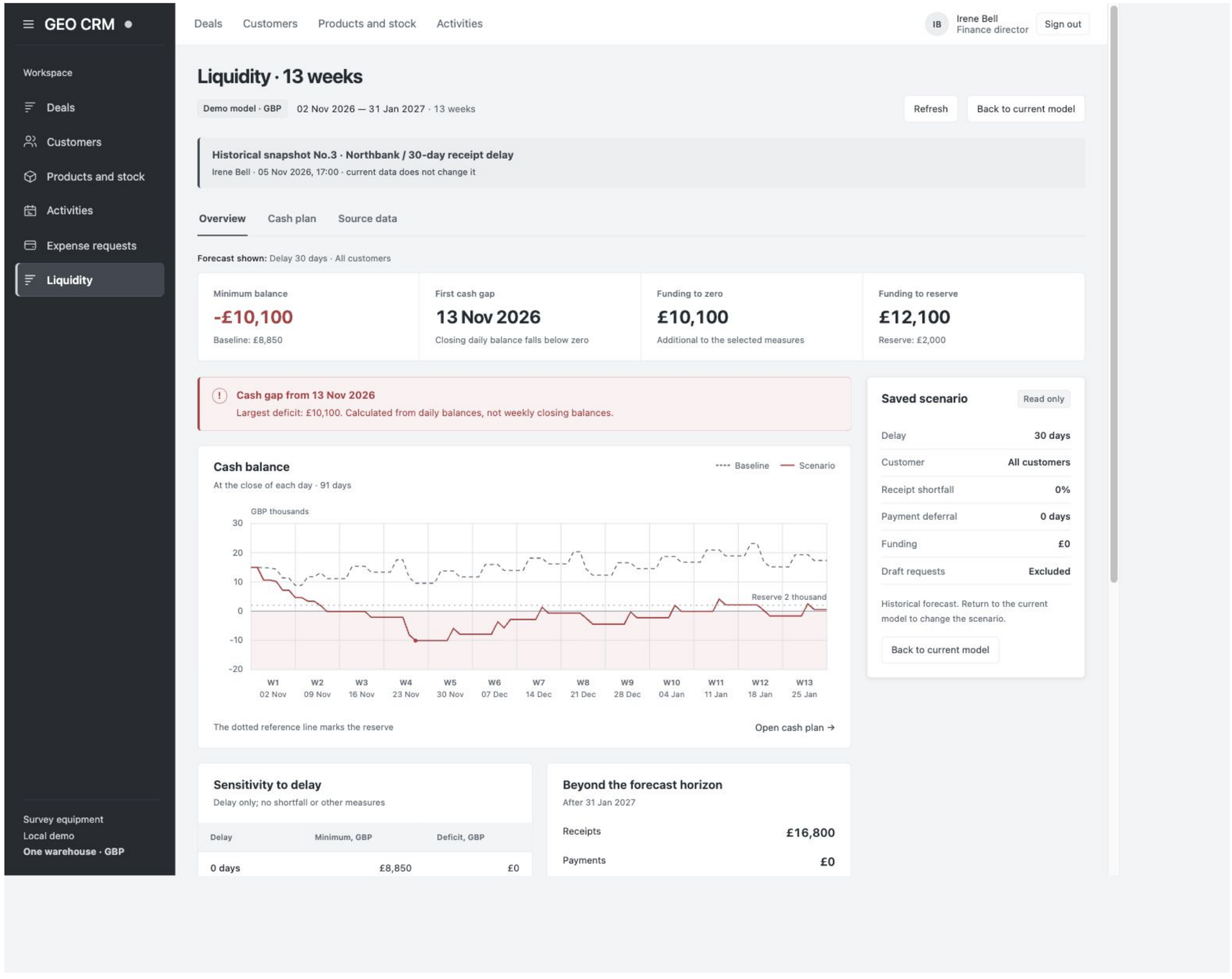


Role: finance director / C10

Finance director: week 11 closes at GBP 2,139 but its minimum daily closing balance is minus GBP 61. Weekly closing balances alone would hide the cash gap. Week 11 is selected in the actual cash-plan interface. The daily table shows minus GBP 61 before the receipt on 14 January, although the week closes at GBP 2,139. Evidence C10 also verifies the underlying rows independently.

PRODUCT IN USE

Figure 11. An inspectable historical result

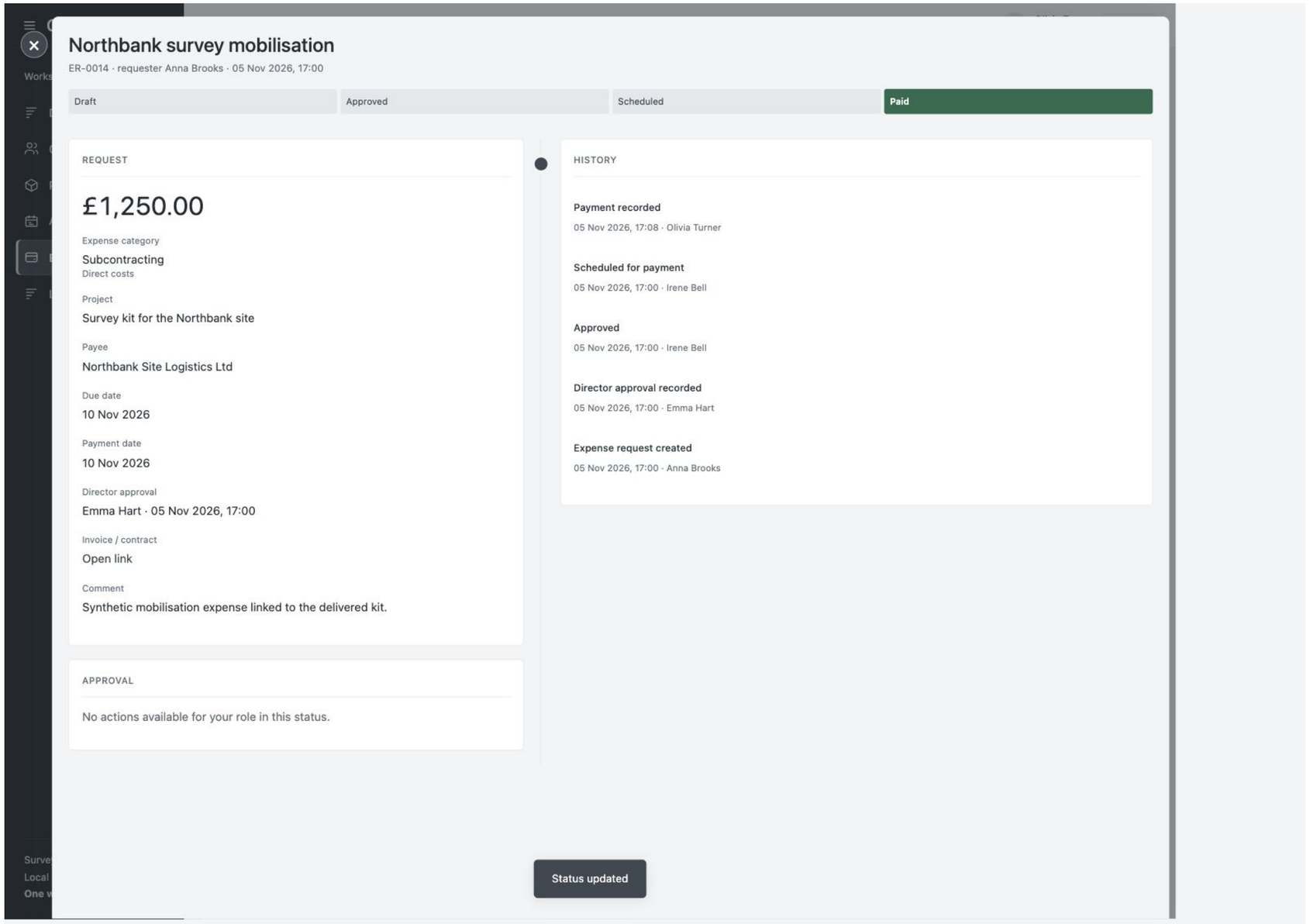


Role: finance director / C11

Finance director: saved forecast 3 preserves the Northbank 30-day scenario, its inputs and results, with author and time. The view is explicitly historical and read-only. Saved forecast 3 stores its inputs and result together. After the accountant later recorded payment, the live open-commitment inputs changed but this snapshot did not. The restart check recovered the same saved inputs and result.

PRODUCT IN USE

Figure 12. Manual payment recording



Role: accountant / C12
Accountant: payment is recorded manually and ER-0014 is marked Paid. This changes the open-commitment forecast input; it does not execute a bank payment or automatically reconcile opening cash. This action was performed in the actual accountant UI, on the isolated demo. The application marks the request Paid and removes it from future open commitments. Opening cash is not automatically reconciled; it must already account for completed transactions when a new forecast is prepared.

ENGINEERING RECORD

8. Fresh checks and their limits

The report used a byte-identical copy of all 30 manifest-listed application files. The original application was left running at localhost:8765. The copied application used HTTP port 8890 and a restored synthetic database on the owned PostgreSQL port 55435. A separate disposable test database and HTTP port 8891 served the unchanged tests; configuration changed only through their supported environment variables.

New check	Observed result	Coverage boundary
Existing acceptance suite	36 passed in 4.54 s.	Actual HTTP/PostgreSQL, including stock concurrency, replay, quotation and role rules, expenses and LIQ-13. This is not exhaustive correctness.
New role-separated case	Completed sales, warehouse, expense and forecast path.	HTTP writes; native UI observations and accountant payment action.
Independent cash arithmetic	182 daily rows, 26 weekly aggregates, 91 funding shifts matched.	Captured synthetic inputs and the selected contrasts; not a validation of forecast realism.
Owned app and database restart	All 19 table contents unchanged; saved model and paid state recovered by HTTP.	Report services only; no crash-recovery or power-loss test.
Preservation of original work	All 19 main table digests and 30 application file hashes unchanged.	The report changed only its disposable database and report/handoff artifacts.

Twelve full native frames and one chart close-up were captured in the in-app browser. Their captions identify the role, action, state and claim. The report did not use generated interface images. The inherited browser zoom was left unchanged; capture metadata records the observed CSS viewport where available, and the original image dimensions are retained.

All historical checks remain attributed to their original releases. The earlier 13-, 22- and 33-test results explain development history; the current 36-test run is the basis for the present suite claim. A previous responsive-width matrix is retained evidence, not newly repeated for this report.

APPENDIX A

Evidence matrix 1 / 2

Every row names the scenario, expected result, observation and basis. Evidence paths are relative to evidence/. The accompanying Markdown/JSON matrix carries the full wording and clickable links.

ID	Function / claim	Scenario	Expected	Observed	Source / check	Evidence
C01	Current product and UI	Load copied release	Identified source and real UI	30 hashes match; real English frames	Manifest + browser	APPLICATION_SHA256.json
C02	Customer/contact links	Create company and contact	Correct company relationship	New company 24; contact 36 linked	HTTP + UI Fig. 2	scenario.json
C03	Deal value and fulfilment	Create lines; ship under role	GBP 2,839; won; reservation empty	Expected total and shipped state	HTTP + UI Fig. 3	scenario.json
C04	Stock and safe replay	Receive 2; reserve 1; ship; replay	On hand 1; reserved 0; replay stable	Both case SKUs match	HTTP + UI Fig. 5	scenario.json
C05	Frozen quotation	Edit deal prices temporarily	Prior quotation unchanged	Snapshot compared; GBP 2,839 retained	HTTP + PDF / Fig. 4	case-quotation.pdf
C06	Expense approval roles	GBP 1,250 request; approve/schedule	Director approval required first	Manager 403; early finance 409; then scheduled	HTTP + UI Fig. 6	scenario-actions.json
C07	Input composition	Read current cash sources	No automatic deal receipt; request once	Explicit receipt; request occurs once in 37 sources	Input + independent count	scenario.json
C08	Zero-delay baseline	Simulate delay 0	Baseline equals scenario	Minimum GBP 8,850; no gap	HTTP + arithmetic / Fig. 8	arithmetic-check.json
C09	30-day stress	Delay all receipts 30 days	Date shift; correct balance	-GBP 10,100; first gap 13 Nov	HTTP + arithmetic / Fig. 9	arithmetic-check.json

APPENDIX A

Evidence matrix 2 / 2

Every row names the scenario, expected result, observation and basis. Evidence paths are relative to evidence/. The accompanying Markdown/JSON matrix carries the full wording and clickable links.

ID	Function / claim	Scenario	Expected	Observed	Source / check	Evidence
C10	Daily/weekly distinction	Inspect week 11	Daily gap not hidden by weekly close	Minimum -GBP 61; close GBP 2,139	Independent weekly check + UI	weekly-arithmetic.csv
C11	Saved forecast stability	Save, pay, restart, reload	Snapshot input/result unchanged	Both remain identical	HTTP + Fig. 11 + restart	restart-check.json
C12	Paid request exclusion	Accountant records payment	Paid absent from open flows	Excluded; no automatic opening-cash update	Real UI + HTTP follow-up	payment-follow-up.json
C13	Current suite scope	Execute unchanged tests	Specified checks pass	36 passed in 4.54 s	New isolated pytest run	current-pytest.txt
C14	Restart persistence	Restart owned app and PostgreSQL	Same table contents and stored state	19 tables identical; HTTP recovery passes	New digest and HTTP checks	restart-check.json
C15	Original work preserved	Compare main source and data	No product/data changes	30 source hashes; 19 tables identical	New before/after digests	original-data-after.json
H01	Historical timing and transfer	Read retained chronology/trace	Attribute intervals and environments	36:11 initial turn; 11:51 DSH turn	Retained evidence; not rerun	historical-record.json
D01	Known finance gaps	Bounded negative probes	Detect access/audit gaps	R01/R02 reproduced; R03 remains policy	New probes after case capture	current-negative-probes.json

ENGINEERING RECORD

9. What remains unresolved

Finding	Current observation	Consequence
R01 / access	A manager denied direct access to another manager's deal can create an expense referring to it and receive the deal title.	A remaining cross-module information-disclosure gap. Reproduced in the disposable copy.
R02 / return reason	The return endpoint accepts an empty reason; a supplied reason is not retained in the record/history.	The return audit trail is incomplete. Reproduced separately from the illustrated request.
R03 / policy	The finance director can create and approve the same request.	A four-eyes rule is not enforced; the policy decision has not been agreed.

These findings are limitations of the current demo, not a conclusion about one model or harness. The report task did not authorise product repair, so no fixes were applied. Negative probes were run after the screenshots and use their own request records; they did not supply the illustrated forecast inputs.

The application remains a local synthetic-data demonstrator: one warehouse, one currency, no serial-number equipment accounting, rentals, repairs, bank execution, accounting integration, uploaded documents or background notifications. The finance model does not establish that its manually assembled cash plan is complete or realistic.

Production deployment, comprehensive security qualification, accessibility conformance, load capacity, multi-OS behaviour and failure recovery have not been established. The Flask development server and local demo authentication are conditions of this demonstration, not a production design recommendation.

The experiment's supported conclusion

Under the observed same-computer conditions, engineering work continued from Codex into DeepSeek Harness and back into Codex on one evolving codebase, with subsequent checks and coherent additions. The folder carried enough actionable context for that observed continuation. There is no controlled comparison establishing that iEWR caused a speed, cost or quality advantage, and no test of arbitrary model/harness interchangeability.

ENGINEERING RECORD

Appendix B. Identity and reproduction

Subject	Recorded identity
Application	GEO CRM 1.3.1-demo; no Git commit/tag in the supplied checkout.
Source revision	1d370fec587d97ba3e8229f87c6e85a1d1f2cf27b9b4a5754271d4df79f0e3b0
Identity algorithm	SHA-256 of sorted path<TAB>file-hash lines, joined by newline without a trailing newline. Thirty source files.
Current iEWR	7.0.0 Eternal Alpha, release-r1; immediate source edition clean-r2. Environment not upgraded.
Active MethodPack	geo-liquidity-v1; 38 active descriptions. The clean base inventory declares 37; the project adds LIQ-13.
Runtime	Python 3.14.6, Flask 3.1.3, psycopg 3.3.6, PostgreSQL 18.4; ReportLab 5.1.0. Exact dependency lock is hashed.
Data profile	Fictional UK/GBP demo; en-GB formatting; explicit 1:100 synthetic redenomination from the earlier RUB data.

The initial and substantial extension work are attributable to the iEWR 7.0.0 line. The exact executed initial environment build was not archived. The current package metadata and active method hashes are recorded independently; the folder name is not used as version evidence.

To repeat the numerical result without any running CRM, run check_arithmetic.py beside scenario.json. To repeat the suite and stateful case, follow evidence/REPRODUCE.md on a separate database and port, with the matching application source. The source copy itself was not altered for this report.

The original 27-page GEO_CRM_REPORT.pdf was found and hashed. It describes the earlier 1.0.0-demo and is not needed to read this report. The previous 25-page case report and later English/theme records are provenance, not substitutes for the fresh execution.

Reproduction boundary. The exact case starts from the retained local English demo snapshot. A fresh seed uses its creation date and can produce different dates and totals. The captured model inputs permit exact arithmetic reproduction even without that database snapshot.

ENGINEERING RECORD

Appendix C. Using the case materials

For the site editor

Use Figure 9 for the quantified liquidity scenario, Figure 3 for the integrated deal and fulfilment context, and Figure 6 for the expense workflow. Figure 1 provides product context. The original JPEG files are included; the PDF is not the only image source. The chart close-up is suitable when week labels must remain readable in a narrow layout.

CASE_SUMMARY_EN.md is a standalone English case description. It distinguishes the observed two-environment continuation from an untested causal claim about iEWR. No site was developed or published as part of this work.

Material	Purpose
REPORT_EN.md / REPORT_EN.pdf	The complete explanation, figures, conclusions and boundaries.
SOURCE_STATE.json / SOURCE_REGISTER.json	Current identity and hashes; origin of historical material; missing information.
CLAIMS.md / CLAIMS.json	Function-to-scenario-to-result links.
scenario.json / scenario-actions.json	Captured inputs, results, record identities and sanitised role-separated action trace.
daily-arithmetic.csv / weekly-arithmetic.csv	The independently checked numerical rows.
current-pytest.txt / current-pytest.xml	Fresh acceptance execution, separate from historical test counts.
REPRODUCE.md / check_arithmetic.py / replay.py	Replay instructions and executable verification routes.
current-negative-probes.json	Specific current limitations, not a hidden amendment to the product.

The transfer package excludes installed dependencies, runtime directories, session keys, database backups, raw private agent sessions and obsolete report copies. Historical source originals remain in the project; English summaries and source digests preserve their role without distributing private context.

Assessment still needed. The human chose the task, policies and visual direction and reported a successful startup after transfer. This current report and the complete business behaviour still require human review; no such acceptance is manufactured here.